

VNDHR: Variational Single Nighttime Image Dehazing for Enhancing Visibility in Intelligent Transportation Systems via Hybrid Regularization

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Abstract—The visibility of images plays a crucial role in Intelligent Transportation Systems (ITS). However, images captured under hazy environments can degrade visual quality, significantly reducing the working performance of ITS. Although existing dehazing methods have achieved remarkable performance for daytime hazy images, they struggle to overcome the unique degradations under nighttime haze conditions such as glows, weak illumination, hidden noise, and color distortions. To simultaneously address these degradations, we propose VNDHR, a novel Variational Nighttime Dehazing framework using Hybrid Regularization focusing on enhancing the perceptual visibility of nighttime hazy scenarios. Specifically, a new physical model that accounts for multiple degradations under nighttime haze conditions is first constructed. Then, a novel hybrid variational model comprising an ℓ_p norm, a weighted ℓ_2 norm, and a total variation regularization is developed to obtain a structure-aware illumination and a noise-free reflectance, simultaneously. To remove the nonhomogeneous haze in the illumination, we employ the dark channel prior to estimate parameters in each grid patch. Furthermore, a simple but effective nonlinear stretching function is designed to enhance the texture in the decomposed reflectance component. Finally, the dehazed illumination and the stretched reflectance are combined to generate a haze-free result. Experiments performed on synthetic and real-world nighttime hazy images prove that our VNDHR framework achieves state-of-the-art dehazing performance, providing results with clear details and less noise. Besides, our VNDHR can also handle various types of degraded

images well, such as low-light images, daytime hazy images, sandstorm images, and underwater images.

Index Terms—Nighttime image dehazing, hybrid regularization, ℓ_p norm, stretching.

I. INTRODUCTION

OUTDOOR vision-based applications, such as intelligent transportation systems (ITS), have brought great convenience to people's daily lives. However, images and videos acquired under nighttime haze conditions usually suffer from poor visibility, glow effects, weak illumination, texture blurring, and noise interference due to the presence of artificial light sources with diverse colors and limited illumination range. Low-quality images not only adversely affect human visual perception but also seriously impact the working performance of outdoor vision-based systems. Furthermore, these degradations also influence subsequent computer vision tasks in ITS, such as vehicle detection and recognition [1], traffic flow detection [2], and traffic object recognition and tracking [3]. Therefore, nighttime haze removal is significantly essential to provide high-quality images for vision-based applications.

The current dehazing algorithms are roughly categorized into two classifications: prior-based methods [4], [5], [6], [7], [8], [9], [10], [11] and learning-based methods [12], [13], [14], [15], [16], [17], [18], [19]. The former presents several effective prior knowledge, such as dark channel prior (DCP) [4], color attenuation prior (CAP) [5], region line prior (RLP) [7], saturation line prior (SLP) [8], and heterogeneous priors (HP) [11], to estimate the parameters and subsequently inverts the atmospheric scattering model for image restoration. On the other hand, the latter method mainly constructs learning-based dehazing neural networks, such as DehazeNet [12], MSCNN [13], PMD-Net [14], Cycle-SNPGAN [15], PA-DC-MSPCNN [16], USCFormer [17], and DCD-Net [19], aiming to build end-to-end mapping relationships from the degraded image to the transmission map or haze-free result. These aforementioned haze removal approaches have achieved good performance for daytime hazy scenarios. Unfortunately, they suffer from significant challenges in effectively restoring nighttime hazy images, as viewed in Fig. 1. This limitation arises because the classical atmospheric scattering model and commonly used priors are

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not well-suited to nighttime hazy scenarios, primarily due to the differing types of degradations involved, such as low illumination, glows, noise, and color shifts. In addition, learning-based dehazing networks often fail to generalize effectively to nighttime conditions.

Compared to the daytime haze environment, the nighttime illumination is characterized by artificial light sources with varying colors, uneven distribution, and limited illumination range. These adverse factors will lead to multiple degradations in the acquired image, including reduced visibility, low light, hidden noise, haze, glows, non-uniform illumination, texture blurring, etc. Consequently, the conventional atmospheric scattering model, widely used in dehazing algorithms, is no longer appropriate for nighttime haze scenes. Therefore, various physical models [20], [21], [22] have been proposed to elucidate the formation of the nighttime hazy image. In addition, researchers have introduced several prior-based algorithms, such as maximum reflectance prior (MRP) [23] and gray haze-line prior (GHLP) [24], to remove the effects of non-uniform illumination and glows. However, these priors may not always be valid for complex real-world nighttime haze scenes. On the other hand, learning-based neural networks [25], [26], [27], [28], [29], [30], [31], [32], [33] have been devised to enhance the visibility of nighttime hazy images. However, since data-driven methods strongly rely on the diversity and volume of training data, which differ from real-world nighttime hazy images in terms of degradation features, they can achieve good visual results on synthetic nighttime hazy images but exhibit poor performance for real-world data. Recently, variation-based nighttime dehazing methods [22], [34], [35], [36] have been developed to leverage integer-order norms (ℓ_0 , ℓ_1 or ℓ_2 norms) to enforce variables within an optimization model. Unfortunately, the current integer-order norms are not able to accurately achieve estimation, potentially resulting in unsatisfactory dehazed results. In summary, while these nighttime haze removal approaches enhance the perceptual visibility of nighttime hazy images, they still have some drawbacks. (1) Previous physical models fail to comprehensively describe the degradations presented in nighttime haze conditions. (2) Existing nighttime dehazing methods overlook several important degradations, such as low light and hidden noise, resulting in insufficient brightness and noise reduction.

To address these challenges, we propose a Variation-based Nighttime Dehazing framework using Hybrid Regularization, called VNDHR. To comprehensively describe the degradations in nighttime haze environments, we combine the atmospheric scattering model with pixel-wise airlight and an improved Retinex model to construct a new physical model that accounts for glows, noise, low light, non-uniform illumination, and haze. Based on the physical model, we propose a unified hybrid variational optimization model for illumination-reflectance decomposition, incorporating ℓ_p , weighted ℓ_2 and ℓ_1 -based total variation regularizations. Then, the illumination component is processed based on the local-wise dark channel prior. In addition, we also devise a simple yet effective nonlinear enhancement function to boost the texture in the reflectance component. Lastly, the dehazed illumination and

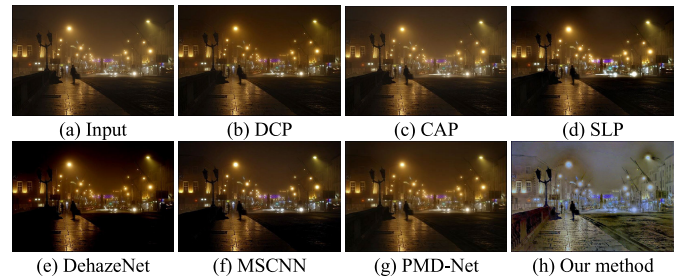


Fig. 1. Daytime dehazing methods are not effective in restoring nighttime hazy images. (a) is an input nighttime hazy image. (b)-(h) are the dehazed results by DCP(TPAMI'11) [4], CAP(TIP'15) [5], SLP(TIP'23) [8], DehazeNet(TIP'16) [12], MSCNN(ECCV'16) [13], PMD-Net(ECCV'22) [14] and our method, respectively.

the stretched reflectance are combined to yield a clear image. The key contributions of this paper are listed as follows.

- We introduce a new physical model that integrates the atmospheric scattering model and Retinex theory to describe multiple degradations presented in nighttime haze environment. Different from previous models [20], [21], [23] that only consider partial degradation factors, our constructed physical model can comprehensively explain the diverse degradations observed in nighttime hazy environment.
- A novel **Hybrid Variational Model (HVM)** consisting of three different regularizations: an ℓ_p norm, a weighted ℓ_2 norm and an ℓ_1 norm, is developed, which can simultaneously achieve a structure-aware illumination and a texture-abundant reflectance with less noise. Compared to previous variational strategies [22], [24], [34], our proposed model offers greater flexibility through the adjustable value of p , allowing it to adapt effectively to the diverse characteristics of real-world scenes.
- A novel **Variational Nighttime image Dehazing** framework using **Hybrid Regularization**, dubbed VNDHR, is proposed to significantly improve the perceptual visibility of nighttime hazy images. In our VNDHR framework, we introduce a local-based DCP and a stretching function that respectively dehaze the illumination and enhance the texture, ultimately yielding a haze-free result. Unlike most existing methods [21], [23], [24] that focus on limited degradation factors, our proposed VNDHR achieves comprehensive improvement by simultaneously addressing glows mitigation, haze removal, illumination compensation, texture enhancement, and noise suppression. This unified approach ensures robust performance across diverse real-world nighttime haze conditions.
- Extensive experiments on real-world and simulated nighttime hazy images demonstrate the superior dehazing performance of our VNDHR framework in terms of visual quality and quantitative assessment. Moreover, our proposed VNDHR framework is capable of processing various types of degraded images, including low-light images, daytime hazy images, sandstorm images, and underwater images.

II. RELATED WORK

The current nighttime dehazing algorithms can be broadly classified into three groups: prior-based methods, learning-based methods, and variation-based methods.

A. Prior-Based Nighttime Dehazing Methods

Pei et al. [37] first use the color transfer technique to adapt the color characteristics of nighttime hazy images into daytime conditions, and then employ the refined DCP and the bilateral filter in local contrast correction to remove haze and enhance visibility. Considering the non-uniform illumination caused by artificial light sources, Zhang et al. [20] propose a novel imaging model that adjusts the constant atmospheric light with point-wise variables for the nighttime haze environment. On the basis of the physical model, a prior named MRP [23] is presented to estimate ambient illumination and transmission for recovering the nighttime hazy image. Li et al. [21] account for the noticeable glow of light sources in the atmospheric scattering model and discover the short tail distribution prior to remove the glow effects. Yu et al. [38] integrate the classic DCP and bright channel prior (BCP) to estimate the transmission map. Lin et al. [39] introduce a near-light source coefficient into the proposed model [20] to describe the glow. In [24], Wang et al. propose GHLP to accurately identify regions with nighttime haze features. However, these developed prior knowledge may not be suitable for complex and diverse real-world scenarios, resulting in subpar dehazing performance.

B. Learning-Based Nighttime Dehazing Methods

Learning-based nighttime dehazing algorithms depend on paired data to automatically learn degradation-relevant features for increasing the visual clarity of nighttime hazy images. Liao et al. [40] propose a haze density prediction network to estimate the density map from an input nighttime hazy image and then subtract this map to obtain a haze-free result. Zhang et al. [25] first present a massive synthetic nighttime hazy image benchmark and then devise an encoder-decoder architecture to achieve nighttime dehazing. Feng et al. [41] design a convolutional autoencoder to estimate the accurate transmission map. To remove the glow in the nighttime scenes, DeGlow-DeHaze [26] is introduced iteratively to capture the relevant features with different receptive fields to estimate the glow term. In [42], generative adversarial network (GAN)-based layer separation method is designed to mitigate the glow effects. Chen et al. [28] propose Ni-DehazeNet using knowledge distillation with neural architecture search. In [43], an end-to-end transformer-based network called NightHazeFormer is developed for supervised pre-training and semi-supervised fine-tuning. Jin et al. [27] leverage guided angular point spread function (GAPSF) and the adaptive gradient convolution to enhance visibility in nighttime hazy images. More recently, Bernabel et al. [30] develop a nighttime dehazing approach to enhance visibility, brighten low-light regions, and suppress glare. A semi-supervised nighttime dehazing network named SFSNiD [31] is proposed to alleviate the haze and glow while realizing natural brightness.

Feng et al. [32] devise a multitype haze degradation model and construct a dual-branch ensemble network framework to achieve unified image dehazing. NightHaze [33] employs masked autoencoder (MAE)-like self-prior learning to increase the clarity of nighttime hazy images. These learning-based nighttime dehazing networks are limited by hardware resources and paired data. Moreover, they struggle to provide satisfactory dehazed results for real-world scenarios because of the disparity between synthetic and real data.

C. Variation-Based Nighttime Dehazing Methods

To reduce computational resources and improve the dehazing performance for real-world scenes, several variation-based algorithms have been proposed to tackle the degradations observed in nighttime haze environment. For instance, He et al. [35] propose a unified framework named SIDE to overcome the issues of both haze removal and illumination enhancement. Yang et al. [34] present a robust variational framework to refine the physically valid initial illumination for edge preservation and halo suppression. Based on Retinex theory, Liu et al. [36] propose a variational decomposition model to achieve nighttime haze removal and enhance textures. In [22], three integer-order norm regularizations are adopted to construct a unified optimization model for layer separation, followed by enhancement and restoration operations for each layer. Although some variational models [44], [45], [46] leverage ℓ_p norm constraints for regularization to enhance low-light images, they are unable to effectively adapt to nighttime hazy scenes and improve the quality of nighttime hazy images, which is attributed to the more intricate degradations in nighttime hazy scenarios. Variation-based dehazing algorithms can achieve superior effectiveness in restoring real-world nighttime hazy images compared to the former two types because they originate from the intrinsic properties of degradation features in real-world scenarios. Thus, the variation-based algorithms are more appropriate for real-world nighttime haze conditions. However, the use of integer-order norm regularization may lead to inaccurate optimization, resulting in unpleasant results.

To address the above issue, we propose a novel hybrid variational model to achieve illumination and reflectance decomposition for nighttime dehazing. Our proposed model integrates a flexible ℓ_p norm, a weighted ℓ_2 norm and an ℓ_1 norm as three regularization terms, which enables it to fully leverage the advantages of both fractional-order and integer-order characteristics. Compared to existing variation-based methods [22], [34], [35], our proposed algorithm provides greater flexibility and is well-suited for handling complex real-world nighttime scenarios.

III. PROPOSED METHOD

In this section, our proposed VNDHR is illustrated in detail. Concretely, we first propose a new physical model to describe multiple degradations. Then, a novel hybrid variational model is constructed to achieve illumination-reflectance decomposition and noise suppression, simultaneously. Finally, the dehazing and enhancement operations are performed on the

illumination and the reflectance, respectively. Fig. 2 depicts the workflow of our proposed VNDHR framework.

A. Physical Model

Since the visible artificial light sources usually introduce noticeable amounts of glow because of multiple scattering, we employ the atmospheric point spread function (APSF) to simulate the glow formation following [21], [27]:

$$G(x) = L(x) \otimes APSF, \quad (1)$$

where x denotes the pixel location, G and L represent the glow and artificial light source terms, respectively, and $\otimes$ denotes the convolution operator.

Low light is also an important factor affecting the perceptual clarity of nighttime hazy images, and the classic Retinex theory is adopted to illustrate the degradation caused by low light. In addition, the hidden noise often exists in the low-light regions and tends to be amplified after enhancement. Therefore, it is essential to consider the hidden noise in the physical model. Many previous methods typically leverage an independent additive term to describe the noise, which is unreasonable for real-world scenes. This is because the illumination component is usually smooth and free of noise, while the reflectance contains more texture and noise [47]. Following [47], the hidden noise is assumed to exist only in the reflectance, and the improved Retinex model is formulated as:

$$S(x) = I(x) \circ (R(x) + N(x)), \quad (2)$$

where ‘ $\circ$ ’ stands for pixel-wise multiplication. S represents the observed degraded image. I , R and N are the illumination, reflectance and noise, respectively.

The illumination I within the improved Retinex model (2) is assumed to be piece-wise smooth and typically includes low-frequency information from the input image S . According to the analysis in [48], the haze often interferes with low-frequency component of degraded image. Therefore, we adopt the atmospheric scattering model and consider non-uniform airlight to construct the haze imaging model for the illumination I :

$$I(x) = J(x) \circ t(x) + A(x) \circ (1 - t(x)), \quad (3)$$

where J , t and A represent the haze-free illumination, medium transmission, and pixel-wise atmospheric light, respectively.

Combining the glow formation model, the improved Retinex model and the atmospheric scattering model, we create a new physical model to describe multiple degradations presented in nighttime haze environment. The constructed physical model is formulated as:

$$\begin{cases} S(x) = I(x) \circ (R(x) + N(x)) + L(x) \otimes APSF \\ I(x) = J(x) \circ t(x) + A(x) \circ (1 - t(x)), \end{cases} \quad (4)$$

Unlike existing physical models for nighttime haze environment, our proposed model can comprehensively elucidate multiple degradations, such as glow, low light, noise, haze and uneven illumination.

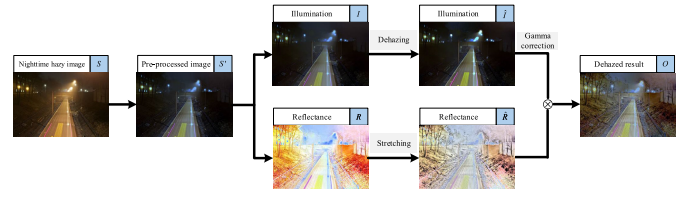


Fig. 2. Flowchart of our proposed VNDHR framework.

B. Hybrid Variational Model

In this part, we propose a novel variational model with hybrid regularization for illumination-reflectance decomposition. To mitigate the adverse effects of glow G prior to decomposition, we first leverage the glow separation strategy [21] as an initial pre-processing phase to acquire a glow-free image S' . Then, a hybrid variational model (HVM) consisting of a flexible ℓ_p norm, a weighted ℓ_2 norm and an ℓ_1 norm is constructed to simultaneously estimate the structure-aware illumination I and the texture-abundant reflectance R with reduced noise by minimizing the following optimization function:

$$E(R, I) = \|R \circ I - S'\|_2^2 + \lambda_1 \|\nabla I\|_p + \lambda_2 W_R \|\nabla R\|_2^2 + \lambda_3 \|\nabla R\|_1 \quad (5)$$

where λ_1 , λ_2 and λ_3 are non-negative parameters for balancing three regularization terms. $\|\cdot\|_p$, $\|\cdot\|_2$, and $\|\cdot\|_1$ represent ℓ_p norm, ℓ_2 norm, and ℓ_1 norm, respectively. W_R is a weight matrix that preserves fine structures in the reflectance. ∇I and ∇R denote the gradients of the illumination and reflectance, respectively. The meaning of each part in the objective function (5) is illustrated as follows:

- $\|R \circ I - S'\|_2^2$ is the data fidelity term that leverages the ℓ_2 norm to guarantee the overall similarity between the estimated $R \circ I$ and S' .
- $\|\nabla I\|_p$ enforces piece-wise smoothness of the illumination to preserve the structural information of S' and prevent the texture-copy issue.
- $W_R \|\nabla R\|_2^2$ encourages fine structures in the reflectance using a weighted ℓ_2 norm, in which the weight matrix combines the Gaussian filter and the traditional total variation [47]. W_R is expressed as follows:

$$W_R = \left(\max(|\nabla R|, \tau) \exp\left((\nabla G_{\sigma_2}(R))^2 / 2\sigma_1^2\right) \right)^{-1}, \quad (6)$$

where G_{σ_2} is the Gaussian filter with spatial width σ_2 and σ_1 is the kernel width of Gaussian function.

- $\|\nabla R\|_1$ constrains the overall smoothness in the reflectance using total variation sparsity to suppress the noise in the estimated R .

Previous variation-based approaches [22], [36] for nighttime dehazing usually rely on integer-order norms, such as ℓ_0 or ℓ_1 norms, to restrain the reflectance, illumination or structural layer within the objective function. However, the use of integer-order norm may lead to insufficient sparse estimation. In contrast to integer-order norms, the fractional-order ℓ_p norm ($0 < p < 1$) can achieve a sparser solution which has proved its effectiveness in several fields such as medical image reconstruction [49]. Therefore, in our study, we employ

ℓ_p norm regularization to encourage the structure-aware illumination component. To the best of our knowledge, our proposed model makes the first attempt to apply ℓ_p norm regularization to nighttime haze removal.

Since our proposed energy function (5) incorporates an ℓ_p norm that makes the optimization problem non-convex, it may be challenging to solve directly. Therefore, a block coordinate descent [50] is employed to search the optimal solution to the non-convex optimization function (5). In addition, we also leverage the re-weighted least square [51] to approximate the ℓ_p and ℓ_1 norms to deal with non-smooth optimization. Consequently, the objective function (5) can be rewritten in the following approximation form:

$$E(R, I) = \|R \circ I - S'\|_2^2 + \lambda_1 G_I \|\nabla I\|_2^2 + \lambda_2 W_R \|\nabla R\|_2^2 + \lambda_3 G_R \|\nabla R\|_2^2, \quad (7)$$

where G_I and G_R represent the weight matrices, and their formulas are defined as follows:

$$G_I = \max(|\nabla I|, \tau_1)^{p-2} \quad (8)$$

$$G_R = \max(|\nabla R|, \tau_2)^{-1} \quad (9)$$

Eq. (7) can be iteratively solved by separating it two sub-problems with respect to I and R . And each sub-problem only includes quadratic terms and thus has a closed-form solution. Here the solutions for the k -th iteration of each sub-problem are provided in detail.

1) Solving the sub-problem of I : We collect the terms associated with I and Eq. (7) is transformed as follows:

$$I_k = \arg \min_I \|R_{k-1} \circ I - S'\|_2^2 + \lambda_1 G_I \|\nabla I\|_2^2. \quad (10)$$

Then, the matrix notation is employed to reformulate Eq. (10) as follows:

$$I_k = \arg \min_I (R_{k-1} \circ I - S')^T (R_{k-1} \circ I - S') + \lambda_1 (I^T D_x^T \tilde{G}_{I_x} D_x I + I^T D_y^T \tilde{G}_{I_y} D_y I), \quad (11)$$

where D_x and D_y represent the Toeplitz matrices from discrete gradient operators with forward difference. $\tilde{G}_{I_x}$ and $\tilde{G}_{I_y}$ are the diagonal matrices including the weights G_I , respectively.

By setting the first-order derivative of Eq. (11) regarding I to zero, we have the following formulation:

$$R_{k-1}^T (R_{k-1} \circ I - S') + \lambda_1 (D_x^T \tilde{G}_{I_x} D_x + D_y^T \tilde{G}_{I_y} D_y) I = 0. \quad (12)$$

The closed-form solution of Eq. (12) can be obtained:

$$I_k = (R_{k-1}^T R_{k-1} + \lambda_1 U)^{-1} (R_{k-1}^T S'), \quad (13)$$

where $U = D_x^T \tilde{G}_{I_x} D_x + D_y^T \tilde{G}_{I_y} D_y$.

2) Solving the sub-problem of R : We omit the terms irrelevant to R and the problem becomes:

$$R_k = \arg \min_R \|R \circ I_k - S'\|_2^2 + \lambda_2 W_R \|\nabla R\|_2^2 + \lambda_3 G_R \|\nabla R\|_2^2. \quad (14)$$

Algorithm 1 Solution of Hybrid Variational Model (5)

Input: pre-processed image S' , parameters $\lambda_1, \lambda_2, \lambda_3, \delta_1, \delta_2, p$, and ε , and maximum number of iterations K .

Output: illumination I and reflectance R

Initialization: $I^0 = S', R^0 = 1$

```

1: for  $k = 1$  to  $K$  do
2:   Compute weights  $G_I$  using Eq. (8).
3:   Update  $I_k$  using Eq. (13).
4:   Compute weights  $G_R$  using Eq. (9).
5:   Compute weights  $W_R$  using Eq. (6).
6:   Update  $R_k$  using Eq. (17).
7:   if  $\|I_k - I_{k-1}\|/\|I_{k-1}\| < \varepsilon$  or
        $\|R_k - R_{k-1}\|/\|R_{k-1}\| < \varepsilon$  then
8:     break
9:   end if
10: end for

```

The matrix form of Eq. (14) is reformulated as:

$$R_k = \arg \min_R (R \circ I_k - S')^T (R \circ I_k - S') + \lambda_2 (R^T D_x^T \tilde{W}_{R_x} D_x R + R^T D_y^T \tilde{W}_{R_y} D_y R) + \lambda_3 (R^T D_x^T \tilde{G}_{R_x} D_x R + R^T D_y^T \tilde{G}_{R_y} D_y R), \quad (15)$$

where $\tilde{W}_{R_{x,y}}$ and $\tilde{G}_{R_{x,y}}$ are the diagonal matrices containing the weights W_R and G_R , respectively.

Similar to the former deduction, we have the first-order derivative of Eq. (15):

$$I_k^T (R \circ I_k - S') + \lambda_2 (D_x^T \tilde{W}_{R_x} D_x + D_y^T \tilde{W}_{R_y} D_y) R + \lambda_3 (D_x^T \tilde{G}_{R_x} D_x + D_y^T \tilde{G}_{R_y} D_y) R = 0. \quad (16)$$

The closed-form solution with regard to R is provided:

$$R_k = (I_k^T I_k + \lambda_2 V + \lambda_3 M)^{-1} (I_k^T S'), \quad (17)$$

where $V = D_x^T \tilde{W}_{R_x} D_x + D_y^T \tilde{W}_{R_y} D_y$ and $M = D_x^T \tilde{G}_{R_x} D_x + D_y^T \tilde{G}_{R_y} D_y$.

To reduce computational complexity, a fast solver called preconditioned conjugate gradient (PCG) [52] is employed to solve Eq. (13) and (17). I and R are updated until the convergence condition $\|I_k - I_{k-1}\|/\|I_{k-1}\| < \varepsilon$ or $\|R_k - R_{k-1}\|/\|R_{k-1}\| < \varepsilon$ is met or the maximum number of iterations K ($K = 10$) is reached.

The iteration process is shown in Fig. 3. As the number of iterations increases, the decomposed illumination becomes progressively smoother while preserving structural edges, and the reflectance component reveals richer textures, which indicates that our proposed HVM can yield structure-aware illumination and texture-abundant reflectance. The entire solution of our proposed HVM is outlined in Algorithm 1.

C. Non-Uniform Dehazing for Illumination

According to previous analysis in our constructed physical model (4), the decomposed illumination is affected by haze

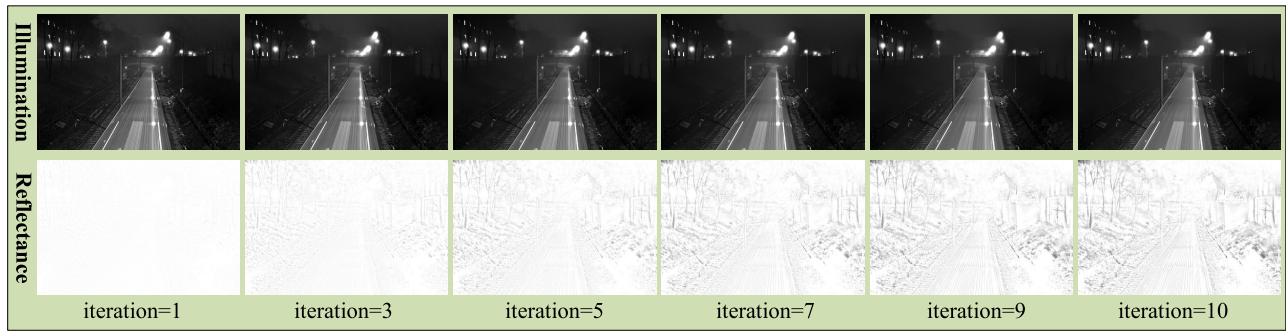


Fig. 3. Iteration process of our proposed hybrid variational model for illumination and reflectance. The first and second row are respectively the generated illumination and reflectance during the iteration process.

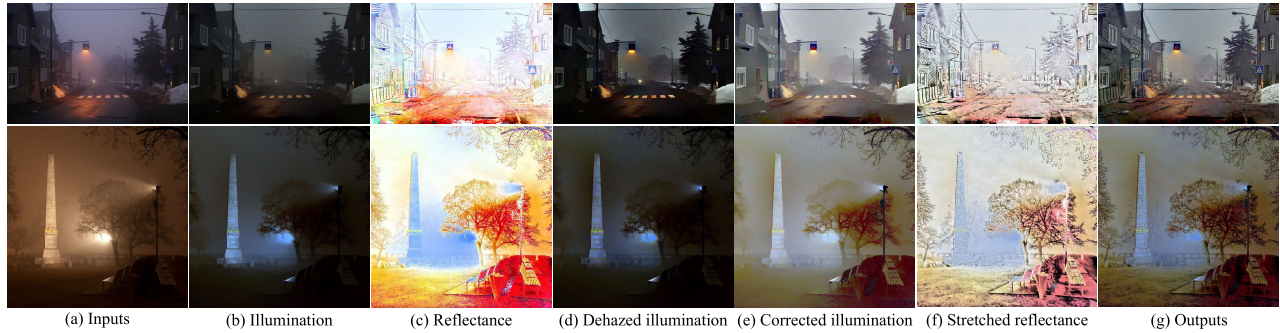


Fig. 4. More examples of the processing procedure of our proposed VNDHR.

interference. Following [21], [22], to mitigate the impact of non-uniform atmospheric light, we improve a classic DCP-based method [4] with non-overlapping local patches to recover the illumination. Specifically, the dark channel of the illumination component is first calculated within each local patch and the top 10 percent brightest pixels in the dark channel are picked. From these pixels, the one with highest intensity is regarded as the atmospheric light for the local patch. Then, the efficient guided filter [53] is leveraged to refine the coarse transmission $t(x)$ and atmospheric light $A(x)$. Afterwards, the haze-free illumination $\hat{I}$ is obtained by inverting the physical model:

$$\hat{I}(x) = \frac{I(x) - A(x)}{\max(t(x), t_0)} + A(x), \quad (18)$$

where t_0 is empirically set to 0.25. By comparing Fig. 4(b) and (d), the haze is effectively removed, and the scene details in the distant regions are recovered, such as the house in the first row of Fig. 4.

D. Nonlinear Stretching for Reflectance

In order to boost the texture in the decomposed reflectance, a simple but effective nonlinear stretching function is designed as follows:

$$\hat{R} = \text{sign}(R) \cdot \left(\frac{255}{1 + e^{\alpha(T-R)}} \right), \quad (19)$$

where $T = \beta \min(\max(R), 255)$, and $\hat{R}$ is the enhanced result. α and β , which are empirically set to 0.05 and 0.85, respectively, control the stretching effect and the threshold.

From Fig. 4(f), the stretched reflectance components significantly enhance the scene details, improving the perceptual visibility of nighttime hazy images.

Finally, the dehazed result O is acquired by combining the dehazed illumination $\hat{I}$ and the stretched reflectance $\hat{R}$:

$$O = \hat{R} \circ \hat{I}^\gamma, \quad (20)$$

where γ is empirically set to 0.35 for brightness compensation. As illustrated in Fig. 4, our proposed VNDHR framework effectively improves the overall brightness, mitigates the glows, removes the haze, enhances the scene details, and suppresses the noise, simultaneously.

IV. EXPERIMENTAL RESULTS

In this section, we initially present the implementation details of our VNDHR framework. Then, the parameters sensitivity experiments are provided. Next, the qualitative and quantitative comparisons are performed to evaluate the effectiveness of our proposed VNDHR. Subsequently, the contributions of each module in our VNDHR framework are demonstrated in the ablation study. Afterwards, the convergence analysis is discussed. Finally, we provide several examples to demonstrate the measurable performance of our algorithm in high-level computer vision tasks and the extended applications of our VNDHR framework.

A. Implementation Details

Our experiments are performed in MATLAB R2019a on a PC with Intel Xeon W-2133 3.6GHz CPU and 32G RAM.

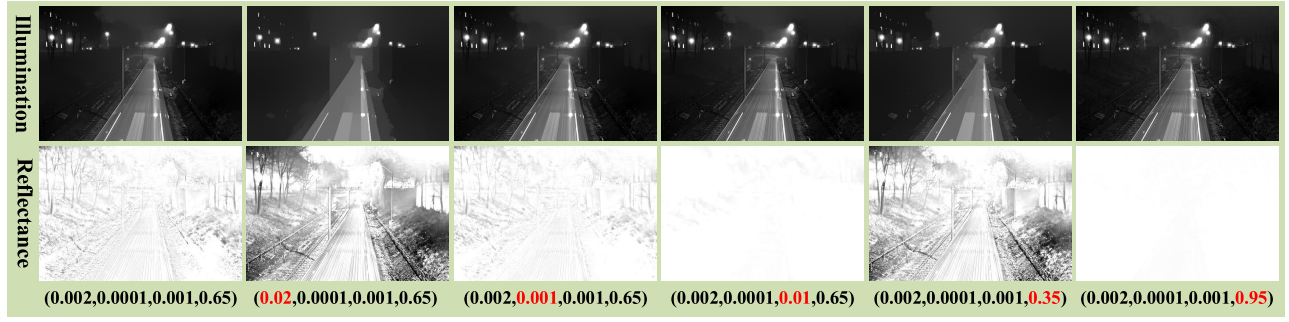


Fig. 5. Effects of parameters (λ_1 , λ_2 , λ_3 , p) for illumination and reflectance. The first column is the estimated illumination and reflectance using our default parameter settings.

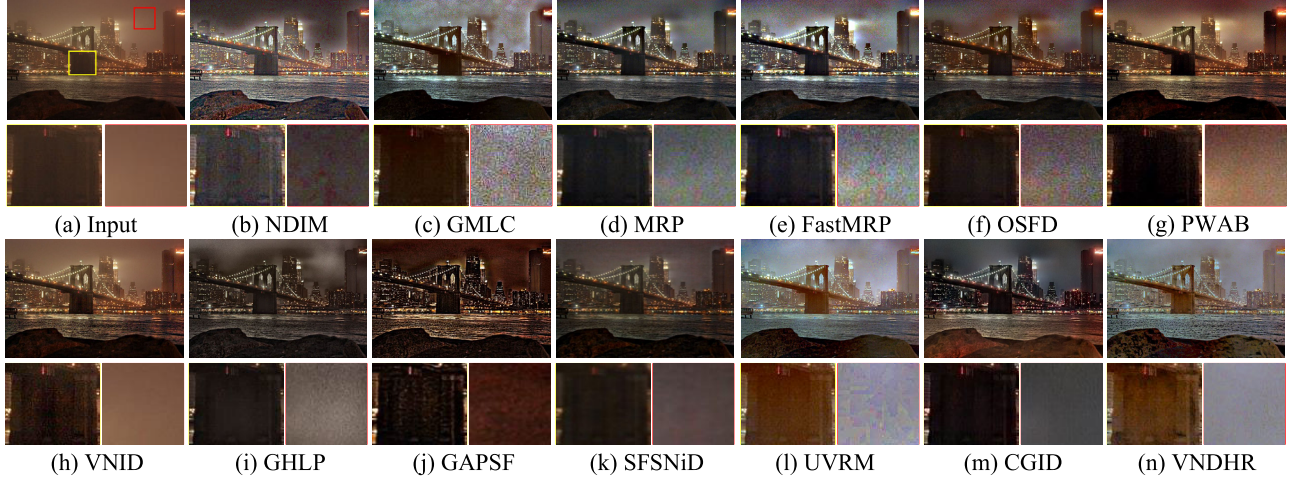


Fig. 6. Comparisons of state-of-the-art nighttime dehazing methods. (a) Input real-world nighttime hazy image. (b)–(n) are the results by NDIM [20], GMLC [21], MRP [23], FastMRP [23], OSFD [25], PWAB [38], VNID [34], GHLP [24], GAPSf [27], SFSNiD [31], UVRM [22], CGID [32], and our VNDHR, respectively. (Zoom in for better visualization.)

If not specifically indicated, the empirical parameters λ_1 , λ_2 , λ_3 , p , δ_1 , δ_2 and ε are set to 0.002, 0.0001, 0.001, 0.65, 1, 1, and 0.001, respectively. Our proposed variational model processes color images in the V-Channel of HSV color space. During the reflectance enhancement phase, the S-Channel is attenuated by multiplying it by a factor less than 1 to prevent oversaturation.

B. Parameters Sensitivity Study

In this subsection, the effects of three regularization parameters (λ_1 , λ_2 , λ_3) and a fractional-order norm parameter p on the illumination and reflectance are investigated. As shown in Fig. 5, the estimated illumination and reflectance in the first column are obtained using our default parameters, while parameters marked with red in the other five columns indicate deviations from the default settings. By analyzing Fig. 5, the estimated illumination with a larger λ_1 ($\lambda_1 = 0.02$) may lead to excessive smoothing of the illumination component, while simultaneously introducing more textures in the reflectance, resulting in the loss of structural information in the illumination and over-enhancement of the reflectance. This is because λ_1 determines the degree of smoothness for the illumination layer. Similarly, the obtained reflectance with a larger λ_2 ($\lambda_2 = 0.001$) exhibits reduced details, which may lead to insufficient enhancement of scene details.

As λ_3 increases ($\lambda_3 = 0.01$), the estimated reflectance will also be excessively smoothed because λ_3 as a regularization parameter controls the overall smoothness of the reflectance to prevent noise amplification. For the flexible p -norm, a smaller value of p ($p = 0.35$) leads to excessive smoothing of the illumination component and an overemphasis on the reflectance details, whereas a larger value of p ($p = 0.95$) fails to effectively maintain the key structural information in the illumination and preserve the details in the reflectance. Generally, our default parameter settings can provide visually pleasing results.

C. Qualitative Comparisons

As depicted in Figs. 6–8, our proposed VNDHR is compared with several specialized nighttime dehazing methods, including NDIM [20], GMLC [21], MRP [23], FastMRP [23], OSFD [25], PWAB [38], VNID [34], GHLP [24], GAPSf [27], SFSNiD [31], UVRM [22], and CGID [32], on real-world and simulated nighttime hazy images. For fair comparisons, the source codes or executable files of NDIM, GMLC, MRP, FastMRP, OSFD, PWAB, VNID, and CGID are obtained from the authors' website and the used parameters in these approaches are set to default values. The dehazed results of GHLP, GAPSf, SFSNiD and UVRM are provided by the authors. From Fig. 6, NDIM, GMLC, MRP, FastMRP, OSFD,

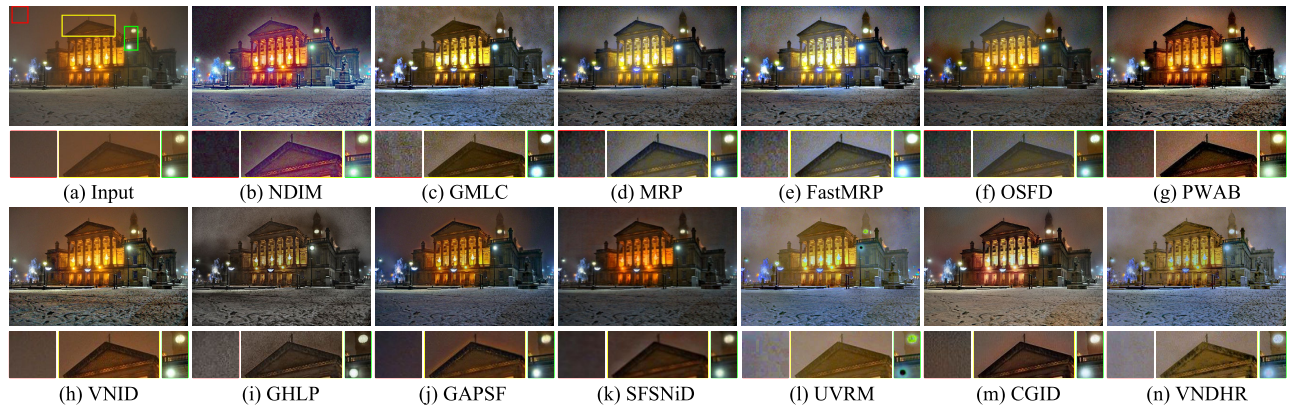


Fig. 7. Comparisons of state-of-the-art nighttime dehazing methods. (a) Input real-world nighttime hazy image. (b)–(n) are the dehazed results by NDIM [20], GMLC [21], MRP [23], FastMRP [23], OSFD [25], PWAB [38], VNID [34], GHLP [24], GAPSf [27], SFSNiD [31], UVRM [22], CGID [32], and our VNDHR, respectively. (Zoom in for better visualization.)

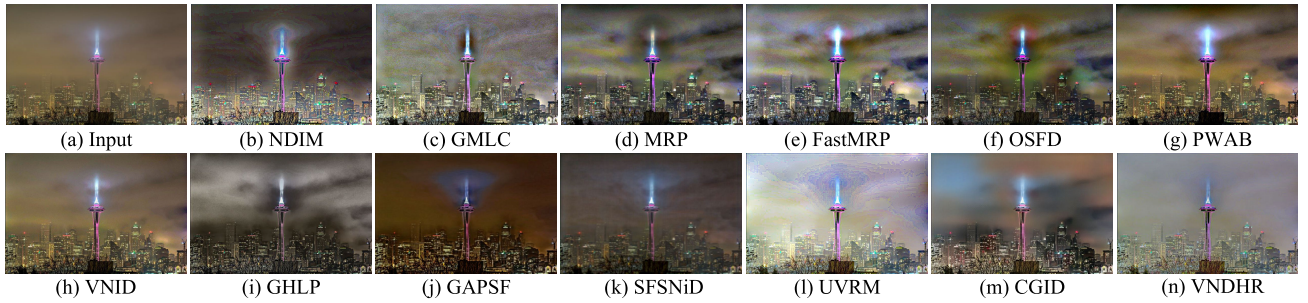


Fig. 8. Comparisons of state-of-the-art nighttime dehazing methods. (a) Input real-world nighttime hazy image. (b)–(n) are the results by NDIM [20], GMLC [21], MRP [23], FastMRP [23], OSFD [25], PWAB [38], VNID [34], GHLP [24], GAPSf [27], SFSNiD [31], UVRM [22], CGID [32], and our VNDHR, respectively. (Zoom in for better visualization.)

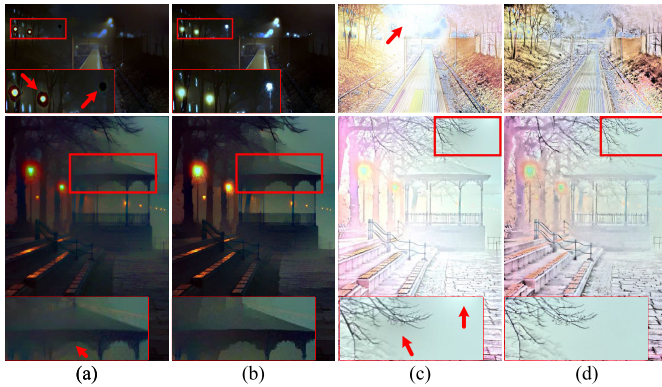


Fig. 9. Advantages of our VNDHR compared to UVRM [22]. (a) and (c) show the dehazed illumination and enhanced reflectance provided by UVRM [22]. (b) and (d) present the dehazed illumination and stretched reflectance generated by our VNDHR, respectively. (Zoom in for better visualization.)

PWAB, and UVRM all introduce artifacts and amplified noise, resulting in unsatisfactory results. VNID, SFSNiD and CGID are unable to suppress the glows around artificial lights. In addition, SFSNiD also leads to the blurred details in the dehazed result. The results of GHLP, GAPSf, and CGID exhibit color shifts, failing to recover the vivid colors. Moreover, these methods struggle to recover the scene details and brightness, such as the enlarged yellow and red rectangles in Fig. 6. As viewed in Fig. 7, the enlarged yellow rectangle

patches illustrate that the compared state-of-the-art nighttime haze removal methods exhibit noticeable halos around depth edges. Moreover, they are also prone to amplifying the hidden noise of dark regions, as seen in the enlarged red patches. Although UVRM can recover the details and brightness, it may introduce severe artifacts around the artificial light sources, as shown in the enlarged green patch in Fig. 7(i). In Fig. 8, other state-of-the-art professional nighttime haze removal methods fail to perform effectively in the sky regions and do not adequately reduce glows and artifacts. Compared to the above algorithms, our VNDHR produces natural results with abundant details, normal brightness, and effective noise and artifacts suppression.

In order to further demonstrate the advantages of our proposed VNDHR, we provide a detailed comparisons with UVRM [22], as shown in Fig. 9. Fig. 9(a) and (c) illustrate the dehazed illumination and enhanced reflectance generated by UVRM. UVRM may lead to noticeable glows and artifacts around the artificial light source, as indicated by the red arrow in the first row of Fig. 9(a). Additionally, the decomposed illumination fails to effectively preserve the structural edges, as seen in the second row of Fig. 9(a). On the other hand, the enhanced reflectance components provided by UVRM introduce halos around depth edges and exhibit oversaturation in the sky regions in Fig. 9(c). These deficiencies arise from the fact that UVRM adopts an incomplete physical model and the integer-order norm constraints for regularization, which

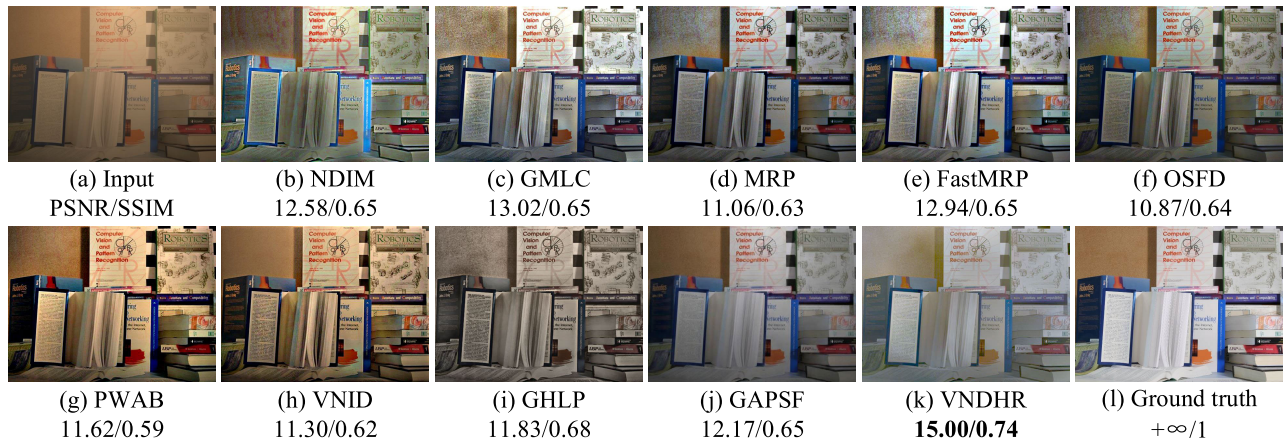


Fig. 10. Comparisons of state-of-the-art nighttime dehazing methods. (a) and (l) are a synthetic nighttime hazy image and the corresponding ground truth selected from NHM dataset. (b)-(k) are the dehazed results by NDIM [20], GMLC [21], MRP [23], FastMRP [23], OSFD [25], PWAB [38], VNID [34], GHLP [24], GAPSf [27], and our VNDHR, respectively.

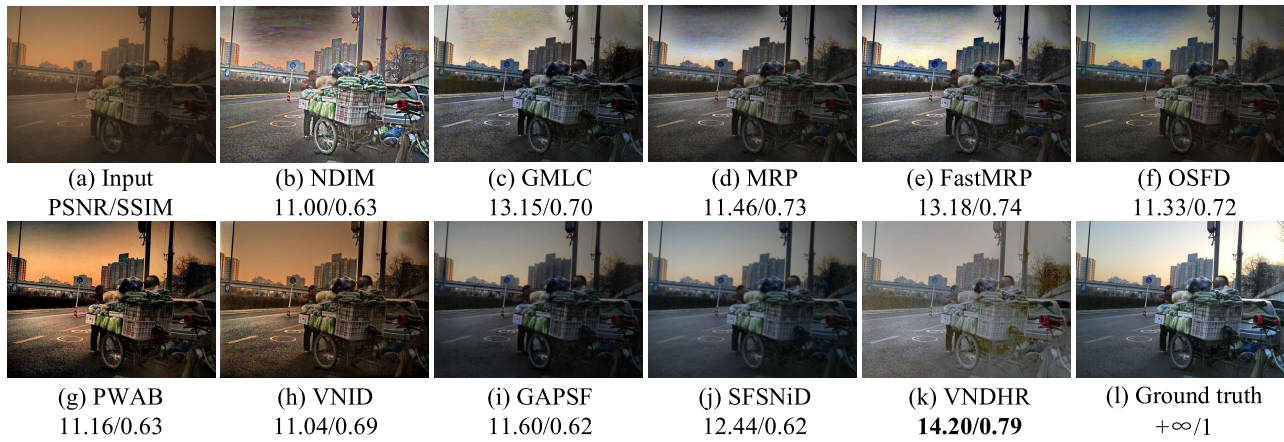


Fig. 11. Comparisons of state-of-the-art nighttime dehazing methods. (a) and (l) are a synthetic nighttime hazy image and the corresponding ground truth selected from NHR dataset. (b)-(k) are the dehazed results by NDIM [20], GMLC [21], MRP [23], FastMRP [23], OSFD [25], PWAB [38], VNID [34], GAPSf [27], SFSNiD [31] and our VNDHR, respectively.

fail to achieve effective decomposition and noise suppression. In Fig. 9(b) and (d), compared to UVRM [22], our proposed algorithm effectively preserves the structural information in the illumination and enhances scene details while simultaneously suppressing halos and artifacts in the reflectance.

The visual comparisons on synthetic nighttime hazy images acquired from the NHM and NHR [25] datasets are presented in Figs. 10-11. From Fig. 10, NDIM, GMLC, MRP, FastMRP, and OSFD introduce amplified noise in the upper left region and also lead to over-enhancement, which compromises the natural appearance of the image. PWAB and VNID struggle with color distortion due to their limited ability to suppress glow effects. GHLP fails to adequately compensate for vivid colors, and the result of GAPSf shows insufficient brightness. In Fig. 11, the results provided by NDIM, MRP, FastMRP, and OSFD exhibit significant distortion in the sky region. GMLC introduces noticeable halo artifacts around regions with depth discontinuities, which further detracts from the visual quality. PWAB and VNID are unable to achieve effective color correction, leading to noticeable color imbalances. Although GAPSf and SFSNiD avoid halos and significant distortions, their results suffer from inadequate brightness, which fails to

fully reveal scene details. In contrast, our VNDHR framework achieves superior visual results with natural brightness, clear details, and reduced noise.

Furthermore, we provide additional real-world and synthetic test images in Fig. 12 to validate the effectiveness of our proposed VNDHR framework. The results demonstrate that VNDHR successfully handles a wide range of degradations, including varying haze densities, diverse color conditions, and low-light scenarios commonly observed in both real-world and simulated nighttime hazy images. As illustrated in Fig. 12, our algorithm not only enhances perceptual visibility by recovering clear structural details and vibrant colors, but also effectively suppresses noise and artifacts. These improvements are critical for applications such as intelligent transportation systems, where enhanced image clarity and accurate feature representation are essential for reliable performance.

D. Quantitative Comparisons

In order to prevent the visual bias, we further demonstrate the effectiveness of our proposed VNDHR on synthetic nighttime hazy image datasets, namely NHM and NHR.

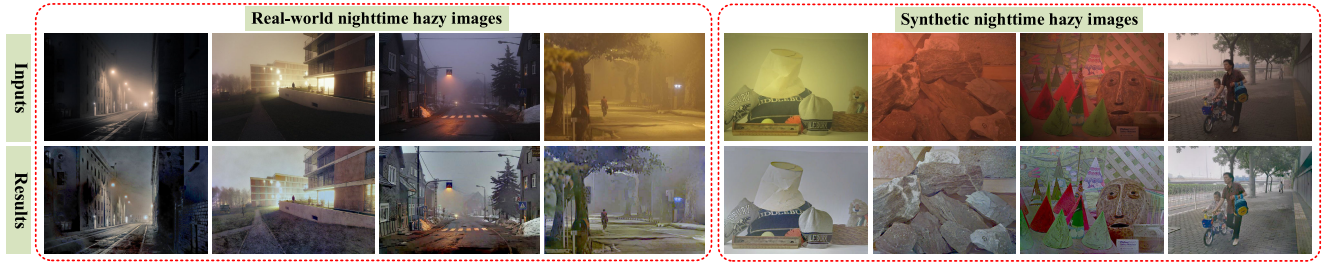


Fig. 12. More visual results on real-world and synthetic nighttime hazy images. The first row is the input real-world and synthetic nighttime hazy images and the second row is the corresponding dehazed results by our proposed VNDHR.

TABLE I
OBJECTIVE COMPARISONS OF STATE-OF-THE-ART NIGHTTIME DEHAZING METHODS ON NHM DATASET

Methods	NDIM [20]	GMLC [21]	MRP [23]	FastMRP [23]	OSFD [25]	PWAB [38]	VNID [34]	GHLP [24]	GAPSF [27]	UVRM [22]	VNDHR
PSNR	12.49	11.90	12.99	13.18	13.30	12.73	12.92	13.15	13.52	13.69	13.87
SSIM	0.58	0.59	0.63	0.62	0.64	0.62	0.63	0.67	0.68	0.68	0.69

TABLE II
OBJECTIVE COMPARISONS OF STATE-OF-THE-ART NIGHTTIME DEHAZING METHODS ON NHR DATASET

Methods	NDIM [20]	GMLC [21]	MRP [23]	FastMRP [23]	OSFD [25]	PWAB [38]	VNID [34]	GAPSF [27]	SFSNiD [31]	VNDHR
PSNR	11.57	13.18	12.09	13.54	11.90	11.69	11.71	13.17	13.59	14.29
SSIM	0.61	0.64	0.70	0.69	0.70	0.64	0.69	0.69	0.66	0.75

TABLE III
ABLATION STUDIES ON EACH MODULE IN OUR VNDHR

Metrics	PSNR	SSIM
without HVM	11.5327	0.5292
without dehazing	13.4496	0.6675
without enhancement	13.7181	0.6063
without hybrid constraints	13.8223	0.6781
our VNDHR	13.8729	0.6902

These two datasets (NHM and NHR) are simulated using the 3R method [25], which respectively comprise 350 and 897 synthetic nighttime hazy images and their corresponding clear and low-light images. We employ clear images as ground truth instead of low-light images, as low-light degradation significantly affects image perceptual visibility. In addition, the fractional-order norm parameter p is set to 0.85 for all synthetic data because less texture is preserved in the reflectance layer to prevent over-enhancement.

Following [22], the synthetic testing images in NHM dataset are all resized to 640×480 for size consistency and fair comparison, and we employ clear images as the reference for PSNR and SSIM calculations, instead of using low-light images. In Table I and II, our VNDHR framework achieves the highest scores for PSNR and SSIM on the NHM and NHR test datasets, verifying the superiority of our algorithm.

E. Ablation Study

The ablation studies are performed on both the synthetic NHM dataset [25] and real-world nighttime hazy images to demonstrate the effectiveness of our contributions. Table III and Fig. 13 provide quantitative results and visual comparisons to illustrate the contribution of each module in our VNDHR

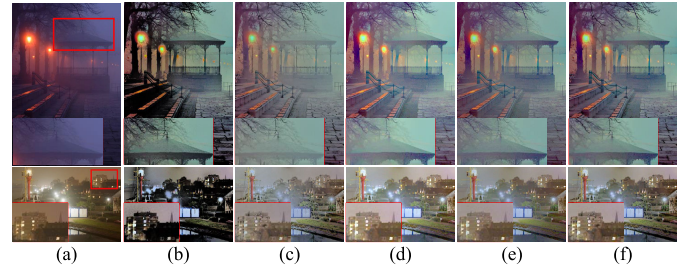


Fig. 13. Ablation studies on each stage of our proposed VNDHR. (a) Input real-world nighttime hazy images. (b) Without our HVM. (c) Without dehazing. (d) Without enhancement. (e) Without hybrid constraints. (f) Our full VNDHR framework. (Zoom in for better visualization.)

framework. Concretely, we first replace our devised hybrid variational model (HVM) with the variational model described in [45] to demonstrate the superiority of our devised HVM. Moreover, the dehazing operation and the stretching module are separately removed from our VNDHR framework to prove the effectiveness of each part. In Table III, our full VNDHR framework achieves highest PSNR and SSIM values compared to other settings. From the visual comparisons in Fig. 13, the dehazed results provided by our full VNDHR framework show more details and less haze. In summary, each module of our VNDHR framework contributes to enhancing the perceptual quality of nighttime hazy images.

Furthermore, to investigate the advantages of our proposed hybrid variational model, we conduct an ablation study comparing our HVM with non-hybrid model that relies solely on ℓ_2 norm constraints for regularization in terms of dehazing performance. More specifically, we set $p = 2$ for the first regularization term (i.e. $\|\nabla I\|_2$) and remove the last ℓ_1 norm constraint on the reflectance ($\|\nabla R\|_1$) to construct

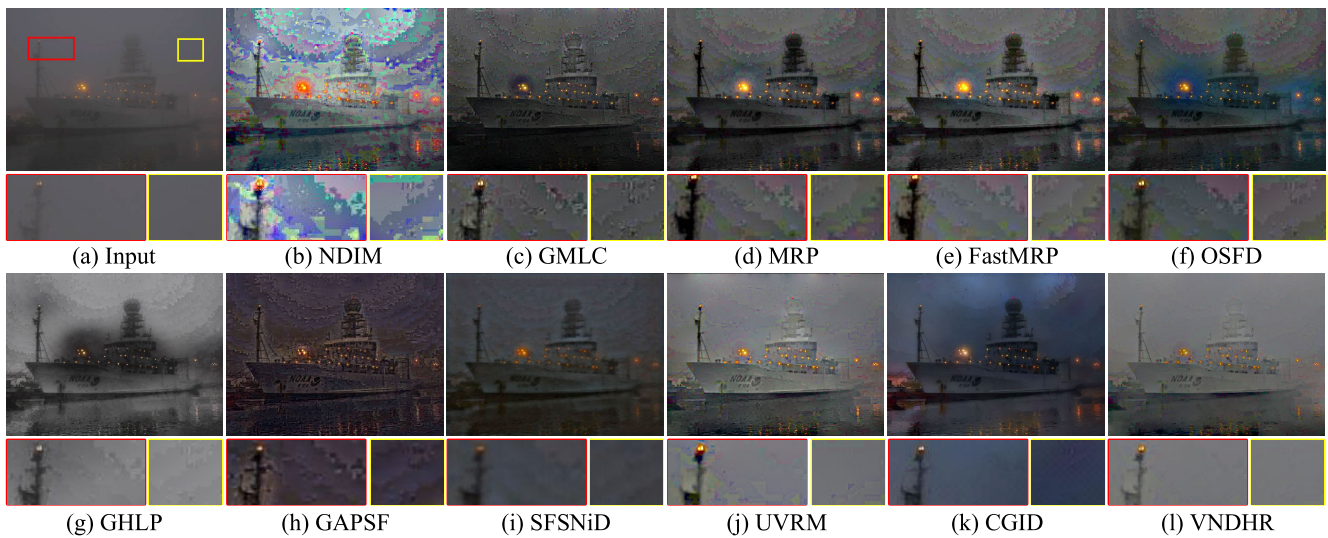


Fig. 14. Comparisons of noise and artifacts suppression for a challenging real-world nighttime hazy image. (a) Input real-world nighttime hazy image. (b)-(l) are the processed results provided by NDIM [20], GMLC [21], MRP [23], FastMRP [23], OSFD [25], GHLP [24], GAPSF [27], SFSNiD [31], UVRM [22], CGID [32] and our VNDHR, respectively. (Zoom in for better visualization.)

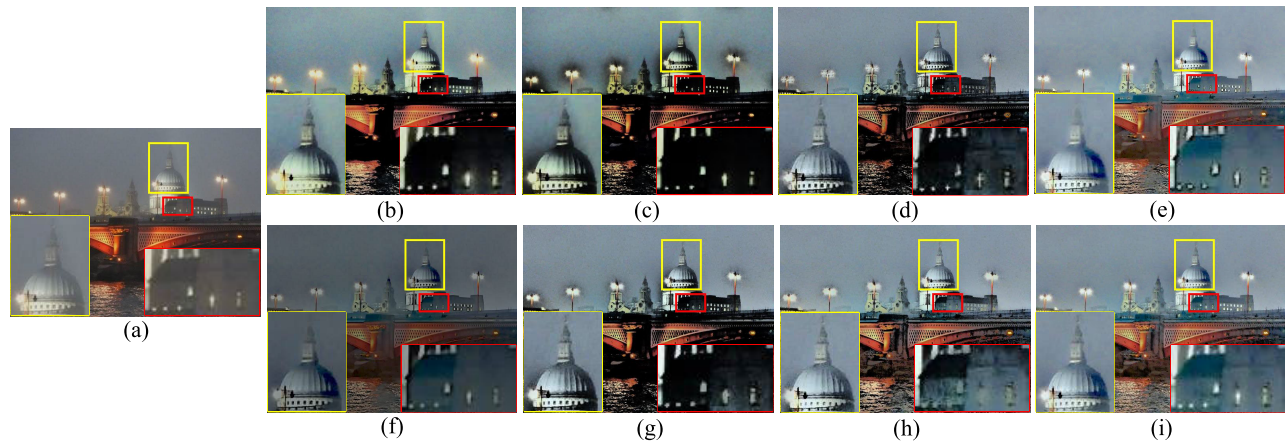


Fig. 15. Comparisons with state-of-the-art variation-based models. (a) Input real-world nighttime hazy image. (b)-(i) are the processed results generated by SIRE [54], WVM [55], JieP [56], SR [57], LR3M [58], SPCV [45], PnPRetinex [46], our VNDHR, respectively.

TABLE IV
OBJECTIVE COMPARISONS OF STATE-OF-THE-ART VARIATIONAL RETINEX MODELS ON NHM DATASET

Metrics	PSNR	SSIM
SIRE [54]	11.5506	0.5784
WVM [55]	11.8574	0.5808
JieP [56]	11.9037	0.5496
SR [57]	13.7970	0.6395
LR3M [58]	13.8000	0.6622
SPCV [45]	11.5327	0.5292
PnPRetinex [46]	13.2388	0.6182
VNDHR	13.8729	0.6902

a non-hybrid variational model, while keeping all other regularization terms and parameters unchanged. In Table III, our proposed VNDHR with hybrid constraints outperforms the non-hybrid model on the NHM dataset. In addition, as shown in the visual comparisons in Fig. 13(e), the non-hybrid model demonstrates limited dehazing capability, attributed to inaccurate decomposition caused by ineffective non-hybrid constraints.

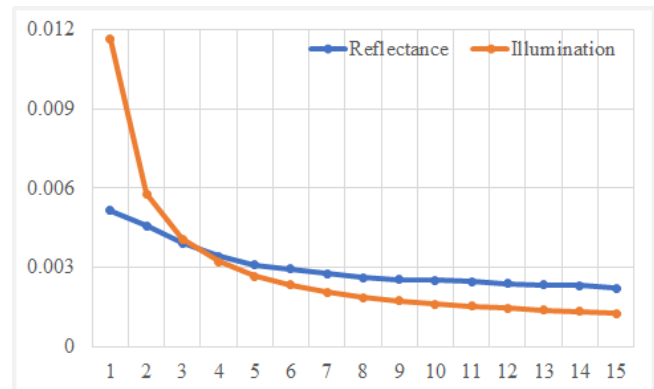


Fig. 16. The average convergence curves of reflectance and illumination on 350 test images from the NHM dataset.

F. Noise and Artifacts Suppression

In order to evaluate the performance of noise and artifacts suppression of our proposed VNDHR, we choose a real-world challenging nighttime hazy images and compare our algorithm

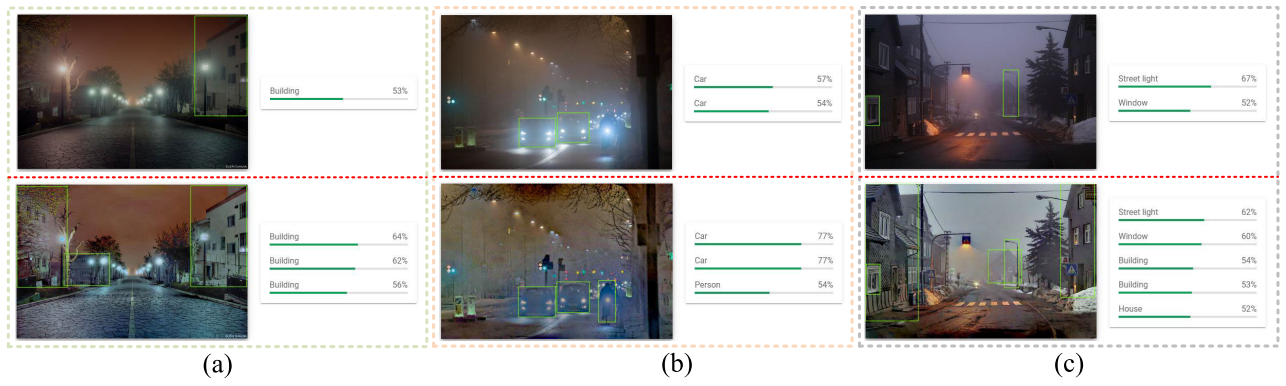


Fig. 17. (a)-(c) represent the measurable performance for high-level vision tasks supported by Google Vision API. The first row is the detection confidence of nighttime hazy images and the second row is the detection confidence of restored results of our VNDHR.

with several state-of-the-art nighttime dehazing approaches. As shown in Fig. 14, most of the compared methods, including NDIM, GMLC, MRP, FastMRP, OSFD, GHLP, GAPSE, SFSNiD, and GCID, suffer from noise amplification and artificial artifacts due to their insufficient to account for the noise interference degradation presented in nighttime hazy scenes. In addition, the dehazed results of GHLP and CGID exhibit color shifts. Although UVRM can suppress some hidden noise to a certain extent, the result by this method is still affected by artificial artifacts and residual noise in the sky regions. This is because UVRM treats the noise as an additive term affecting the entire image, rather than isolating it within the reflectance component. In contrast, our proposed VNDHR achieves superior dehazing result, effectively reducing both noise and artifacts while recovering scene details.

G. Comparisons With State-of-the-Art Variation-Based Models

In order to further validate the superiority of our proposed HVM, we conduct a comparative analysis with several state-of-the-art Retinex-based variational models: SIRE [54], WVM [55], JieP [56], SR [57], LR3M [58], SPCV [45], and PnPRetinex [46], on a synthetic dataset NHM. These variation-based models are adopted to replace our devised HVM in our VNDHR framework while other modules remain unchanged. The parameters used in these models are set to their optimal values as described in the respective literature. Table IV reveals that our proposed HVM outperforms other well-known models for nighttime haze removal.

Moreover, the visual comparison of these presented state-of-the-art variation-based models on a real-world nighttime hazy image is provided in Fig. 15. The dehazed results generated by SIRE, WVM, JieP, SPCV, and PnPRetinex are prone to over-enhancement. The result of LR3M tends to exhibit insufficient brightness. Although SR provides a relatively good result, this method suffers from excessive smoothness and inadequate enhancement for texture. Compared to the aforementioned models, our proposed HVM achieves an ideal illumination-reflectance decomposition and produces a visually superior result with rich details and enhanced visibility, demonstrating the superiority of our algorithm.

H. Convergence Analysis of Hybrid Variational Model

Fig. 16 displays the average convergence curves over 15 iterations for all 350 test images in the NHM dataset. We observe that the estimated illumination and reflectance using our proposed HVM converges quickly within 10 iterations. Therefore, setting the maximum iteration to 10 is sufficient for producing excellent dehazed results.

I. Computational Complexity

Our proposed algorithm consists of four main components: pre-processing, HVM optimization, non-uniform dehazing, and nonlinear stretching. In our framework, the HVM utilizes the PCG method [52] as a fast solver, achieving a computational complexity of $O(N)$ for each iteration, where N is the number of pixels. However, the computational time may vary depending on the number of iterations. Therefore, to provide an intuitive understanding of the computational complexity of our proposed algorithm, the execution time of each module is presented. Processing a test image with a resolution of 575×382 (Fig. 2) takes approximately 35.69 seconds, with each module requiring 2.80, 6.60, 26.27, and 0.02 seconds, respectively. The time-consuming reason for the non-uniform dehazing stems from the patch-based parameter estimation operation. The running speed of our algorithm can be improved using C/C++, as it is currently implemented in MATLAB without optimization. In future work, we will further optimize the computational complexity and develop an efficient algorithm for nighttime haze removal.

J. Performance for High-Level Vision Tasks

In order to verify the effectiveness of our VNDHR framework for high-level computer vision tasks, we employ Google Vision API¹ to show the measurable performance for vision-based tasks. Fig. 17 presents the detection confidence of the nighttime hazy images and the corresponding restored results of our VNDHR, supported by Google Vision API. From Fig. 17, the degradations presented under nighttime haze conditions may limit the capabilities of high-level computer vision tasks, while our proposed VNDHR

¹Google Vision API: <https://cloud.google.com/vision/>

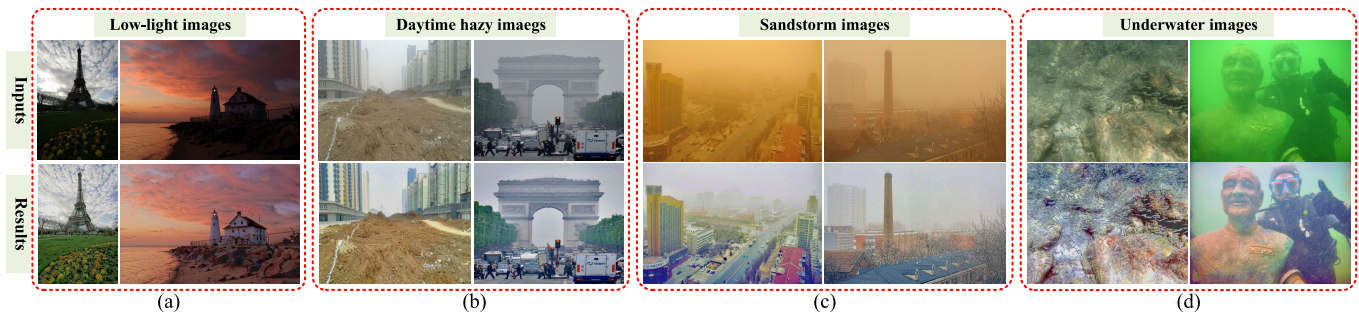


Fig. 18. Visual results of our VNDHR framework on various types of deteriorated images. (a)-(d) are low-light images, daytime hazy images, sandstorm images, and underwater images, respectively. The first row is the input degraded images, and the second row is the restored results of our VNDHR.

framework effectively enhances the perceptual visibility and quality of input nighttime hazy images. Therefore, the results obtained after processing with our VNDHR framework can significantly facilitate the advancement of high-level tasks.

K. Extended Applications

Our proposed VNDHR, with minor modifications, can effectively address other similar low-level computer vision tasks, improving the perceptual visibility of various degraded scenarios, such as low-light images, daytime hazy images, sandstorm images, and underwater images. It is worth noting that our algorithm extends to these vision-based tasks without any parameter adjustment. From Fig. 18, our VNDHR framework can produce visually appealing results with natural illumination, fine details, and corrected colors for various types of degraded images.

V. CONCLUSION

In this paper, we propose a variational nighttime dehazing framework using hybrid regularization named VNDHR. First, we introduce a Retinex-based physical model to interpret the degradations of nighttime haze scenarios. Relying on the physical model, we further develop a novel variational model consisting of three hybrid regularizations: an ℓ_p norm, a weighted ℓ_2 norm and an ℓ_1 norm, to estimate a structure-aware illumination and a noise-free reflectance, simultaneously. Then, the dehazing and enhancement techniques are devised to respectively deal with the illumination and reflectance. Finally, the restored result is acquired by integrating the dehazed illumination and the stretched reflectance. Experiments on real-world and synthetic data demonstrate the superiority of our proposed VNDHR. Furthermore, our framework is capable of promoting high-level tasks and handling low-light images, daytime hazy images, sandstorm images, and underwater images.

REFERENCES

- [1] X. Hu et al., "SINet: A scale-insensitive convolutional neural network for fast vehicle detection," *IEEE Trans. Intell. Transp. Syst.*, vol. 20, no. 3, pp. 1010–1019, Mar. 2019.
- [2] C. Chen, B. Liu, S. Wan, P. Qiao, and Q. Pei, "An edge traffic flow detection scheme based on deep learning in an intelligent transportation system," *IEEE Trans. Intell. Transp. Syst.*, vol. 22, no. 3, pp. 1840–1852, Mar. 2021.
- [3] Y. Zhang, B. Song, X. Du, and M. Guizani, "Vehicle tracking using surveillance with multimodal data fusion," *IEEE Trans. Intell. Transp. Syst.*, vol. 19, no. 7, pp. 2353–2361, Jul. 2018.
- [4] K. He, J. Sun, and X. Tang, "Single image haze removal using dark channel prior," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 33, no. 12, pp. 2341–2353, Dec. 2011.
- [5] Q. Zhu, J. Mai, and L. Shao, "A fast single image haze removal algorithm using color attenuation prior," *IEEE Trans. Image Process.*, vol. 24, no. 11, pp. 3522–3533, Nov. 2015.
- [6] M. Ju, C. Ding, Y. J. Guo, and D. Zhang, "IDGCP: Image dehazing based on gamma correction prior," *IEEE Trans. Image Process.*, vol. 29, pp. 3104–3118, 2020.
- [7] M. Ju, C. Ding, C. A. Guo, W. Ren, and D. Tao, "IDRLP: Image dehazing using region line prior," *IEEE Trans. Image Process.*, vol. 30, pp. 9043–9057, 2021.
- [8] P. Ling, H. Chen, X. Tan, Y. Jin, and E. Chen, "Single image dehazing using saturation line prior," *IEEE Trans. Image Process.*, vol. 32, pp. 3238–3253, 2023.
- [9] G. Sahu, A. Seal, O. Krejcar, and A. Yazidi, "Single image dehazing using a new color channel," *J. Vis. Commun. Image Represent.*, vol. 74, Jan. 2021, Art. no. 103008.
- [10] Z. Qiu et al., "Perception-oriented UAV image dehazing based on super-pixel scene prior," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5913519.
- [11] S. Liang, T. Gao, T. Chen, and P. Cheng, "A remote sensing image dehazing method based on heterogeneous priors," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5619513.
- [12] B. Cai, X. Xu, K. Jia, C. Qing, and D. Tao, "DehazeNet: An end-to-end system for single image haze removal," *IEEE Trans. Image Process.*, vol. 25, no. 11, pp. 5187–5198, Nov. 2016.
- [13] W. Ren, S. Liu, H. Zhang, J. Pan, X. Cao, and M.-H. Yang, "Single image dehazing via multi-scale convolutional neural networks," in *Proc. Eur. Conf. Comput. Vis.* Springer, 2016, pp. 154–169.
- [14] T. Ye et al., "Perceiving and modeling density for image dehazing," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*. Cham, Switzerland: Springer, 2022, pp. 130–145.
- [15] Y. Wang et al., "Cycle-SNSPGAN: Towards real-world image dehazing via cycle spectral normalized soft likelihood estimation patch GAN," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 11, pp. 20368–20382, Nov. 2022.
- [16] G. Sahu, A. Seal, D. Bhattacharjee, R. Frischer, and O. Krejcar, "A novel parameter adaptive dual channel MSPCNN based single image dehazing for intelligent transportation systems," *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 3, pp. 3027–3047, Mar. 2023.
- [17] Y. Wang, J. Xiong, X. Yan, and M. Wei, "USCFormer: Unified transformer with semantically contrastive learning for image dehazing," *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 10, pp. 11321–11333, Oct. 2023.
- [18] Y. Feng, L. Ma, X. Meng, F. Zhou, R. Liu, and Z. Su, "Advancing real-world image dehazing: Perspective, modules, and training," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 12, pp. 9303–9320, Dec. 2024.
- [19] G. Sahu, A. Seal, A. Yazidi, and O. Krejcar, "A dual-channel dehaze-net for single image dehazing in visual Internet of Things using PYNQ-Z2 board," *IEEE Trans. Autom. Sci. Eng.*, vol. 21, no. 1, pp. 305–319, Jan. 2024.
- [20] J. Zhang, Y. Cao, and Z. Wang, "Nighttime haze removal based on a new imaging model," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2014, pp. 4557–4561.

- [21] Y. Li, R. T. Tan, and M. S. Brown, "Nighttime haze removal with glow and multiple light colors," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 226–234.
- [22] Y. Liu, Z. Yan, J. Tan, and Y. Li, "Multi-purpose oriented single nighttime image haze removal based on unified variational Retinex model," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 33, no. 4, pp. 1643–1657, Apr. 2023.
- [23] J. Zhang, Y. Cao, S. Fang, Y. Kang, and C. W. Chen, "Fast haze removal for nighttime image using maximum reflectance prior," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jul. 2017, pp. 7418–7426.
- [24] W. Wang, A. Wang, and C. Liu, "Variational single nighttime image haze removal with a gray haze-line prior," *IEEE Trans. Image Process.*, vol. 31, pp. 1349–1363, 2022.
- [25] J. Zhang, Y. Cao, Z.-J. Zha, and D. Tao, "Nighttime dehazing with a synthetic benchmark," in *Proc. 28th ACM Int. Conf. Multimedia*, Oct. 2020, pp. 2355–2363.
- [26] S. Kuanar, D. Mahapatra, M. Bilas, and K. R. Rao, "Multi-path dilated convolution network for haze and glow removal in nighttime images," *Vis. Comput.*, vol. 38, pp. 1121–1134, Mar. 2022.
- [27] Y. Jin, B. Lin, W. Yan, Y. Yuan, W. Ye, and R. T. Tan, "Enhancing visibility in nighttime haze images using guided APSF and gradient adaptive convolution," in *Proc. 31st ACM Int. Conf. Multimedia*, Oct. 2023, pp. 2446–2457.
- [28] H. Chen, N. Li, and R. Chen, "Ni-DehazeNet: Representation learning via bilevel optimized architecture search for nighttime dehazing," *Vis. Comput.*, vol. 40, pp. 6155–6170, Nov. 2023.
- [29] H. Chen, R. Chen, L. Ma, and N. Li, "Single-image dehazing via depth-guided deep Retinex decomposition," *Vis. Comput.*, vol. 39, pp. 5279–5291, Nov. 2023.
- [30] S. A. Bernabel and S. S. Agaian, "NDELS: A novel approach for nighttime dehazing, low-light enhancement, and light suppression," *IEEE Trans. Multimedia*, vol. 26, pp. 9292–9303, 2024.
- [31] X. Cong, J. Gui, J. Zhang, J. Hou, and H. Shen, "A semi-supervised nighttime dehazing baseline with spatial-frequency aware and realistic brightness constraint," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 2631–2640.
- [32] Y. Feng, Z. Su, L. Ma, X. Li, R. Liu, and F. Zhou, "Bridging the gap between haze scenarios: A unified image dehazing model," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 11, pp. 11070–11085, Nov. 2024.
- [33] B. Lin, Y. Jin, W. Yan, W. Ye, Y. Yuan, and R. T. Tan, "NightHaze: Nighttime image dehazing via self-prior learning," 2024, *arXiv:2403.07408*.
- [34] C.-H. Yang, Y.-H. Lin, and Y.-C. Lu, "A variation-based nighttime image dehazing flow with a physically valid illumination estimator and a luminance-guided coloring model," *IEEE Access*, vol. 10, pp. 50153–50166, 2022.
- [35] R. He, X. Guo, and Z. Shi, "SIDE—A unified framework for simultaneously dehazing and enhancement of nighttime hazy images," *Sensors*, vol. 20, no. 18, p. 5300, Sep. 2020.
- [36] Y. Liu, Z. Yan, A. Wu, T. Ye, and Y. Li, "Nighttime image dehazing based on variational decomposition model," in *Proc. IEEE/CVF Conf. on Comput. Vis. Pattern Recognit. Workshops*, Jun. 2022, pp. 640–649.
- [37] S.-C. Pei and T.-Y. Lee, "Nighttime haze removal using color transfer pre-processing and dark channel prior," in *Proc. 19th IEEE Int. Conf. Image Process.*, Sep. 2012, pp. 957–960.
- [38] T. Yu, K. Song, P. Miao, G. Yang, H. Yang, and C. Chen, "Nighttime single image dehazing via pixel-wise alpha blending," *IEEE Access*, vol. 7, pp. 114619–114630, 2019.
- [39] S. Lin, P. Sun, H. Gao, and Z. Ju, "Haze optical-model-based nighttime image dehazing by modifying attenuation and atmospheric light," *J. Opt. Soc. Am. A*, vol. 39, no. 10, pp. 1893–1902, 2022.
- [40] Y. Liao, Z. Su, X. Liang, and B. Qiu, "HDP-Net: Haze density prediction network for nighttime dehazing," in *Proc. Pacific Rim Conf. Multimedia (PCM)*. Cham, Switzerland: Springer, Jan. 2018, pp. 469–480.
- [41] M. Feng, T. Yu, M. Jing, and G. Yang, "Learning a convolutional autoencoder for nighttime image dehazing," *Information*, vol. 11, no. 9, p. 424, Aug. 2020.
- [42] B. Koo and G. Kim, "Nighttime haze removal with glow decomposition using GAN," in *Proc. Asian Conf. Pattern Recognit. (ACPR)*. Cham, Switzerland: Springer, 2020, pp. 807–820.
- [43] Y. Liu, Z. Yan, S. Chen, T. Ye, W. Ren, and E. Chen, "NightHazeFormer: Single nighttime haze removal using prior query transformer," in *Proc. 31st ACM Int. Conf. Multimedia*, Oct. 2023, pp. 4119–4128.
- [44] G. Fu, L. Duan, and C. Xiao, "A hybrid L₂-L_p variational model for single low-light image enhancement with bright channel prior," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Sep. 2019, pp. 1925–1929.
- [45] M. Zhou, X. Wu, X. Wei, T. Xiang, B. Fang, and S. Kwong, "Low-light enhancement method based on a Retinex model for structure preservation," *IEEE Trans. Multimedia*, vol. 26, pp. 650–662, 2023.
- [46] Y.-H. Lin and Y.-C. Lu, "Low-light enhancement using a plug-and-play Retinex model with shrinkage mapping for illumination estimation," *IEEE Trans. Image Process.*, vol. 31, pp. 4897–4908, 2022.
- [47] S. Hao, X. Han, Y. Guo, X. Xu, and M. Wang, "Low-light image enhancement with semi-decoupled decomposition," *IEEE Trans. Multimedia*, vol. 22, no. 12, pp. 3025–3038, Dec. 2020.
- [48] M. Ju, Z. Gu, and D. Zhang, "Single image haze removal based on the improved atmospheric scattering model," *Neurocomputing*, vol. 260, pp. 180–191, Oct. 2017.
- [49] E. Y. Sidky, R. Chartrand, J. M. Boone, and X. Pan, "Constrained T_pV minimization for enhanced exploitation of gradient sparsity: Application to CT image reconstruction," *IEEE J. Transl. Eng. Health Med.*, vol. 2, 2014, Art. no. 1800418.
- [50] P. Tseng, "Convergence of a block coordinate descent method for nondifferentiable minimization," *J. Optim. Theory Appl.*, vol. 109, no. 3, pp. 475–494, Jun. 2001.
- [51] E. J. Candes, M. B. Wakin, and S. P. Boyd, "Enhancing sparsity by reweighted ℓ_1 minimization," *J. Fourier Anal. Appl.*, vol. 14, no. 5, pp. 877–905, 2008.
- [52] R. Barrett et al., *Templates for the Solution of Linear Systems: Building Blocks for Iterative Methods*. Philadelphia, PA, USA: SIAM, 1994.
- [53] K. He, J. Sun, and X. Tang, "Guided image filtering," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 35, no. 6, pp. 1397–1409, Jun. 2013.
- [54] X. Fu, Y. Liao, D. Zeng, Y. Huang, X.-P. Zhang, and X. Ding, "A probabilistic method for image enhancement with simultaneous illumination and reflectance estimation," *IEEE Trans. Image Process.*, vol. 24, no. 12, pp. 4965–4977, Dec. 2015.
- [55] X. Fu, D. Zeng, Y. Huang, X.-P. Zhang, and X. Ding, "A weighted variational model for simultaneous reflectance and illumination estimation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 2782–2790.
- [56] B. Cai, X. Xu, K. Guo, K. Jia, B. Hu, and D. Tao, "A joint intrinsic-extrinsic prior model for Retinex," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 4000–4009.
- [57] M. Li, J. Liu, W. Yang, X. Sun, and Z. Guo, "Structure-revealing low-light image enhancement via robust Retinex model," *IEEE Trans. Image Process.*, vol. 27, no. 6, pp. 2828–2841, Jun. 2018.
- [58] X. Ren, W. Yang, W. Cheng, and J. Liu, "LR3M: Robust low-light enhancement via low-rank regularized Retinex model," *IEEE Trans. Image Process.*, vol. 29, pp. 5862–5876, 2020.



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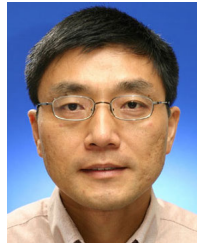
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